

COSE461 Spring 2026 Assignment 2

Word2Vec and Dependency Parsing

Due date: March 27 at 11:59 (KST)

In this assignment, you will review the mathematics behind Word2Vec and build a neural dependency parser using PyTorch. In Part 1, you will explore the partial derivatives involved in training a Word2vec model using the naive softmax loss. In Part 2, you will learn about two general neural network techniques (Adam Optimization and Dropout). In Part 3, you will implement and train a dependency parser using the techniques from Part 2, before analyzing a few erroneous dependency parses.

1. Understanding word2vec (20 points)

Recall that the key insight behind word2vec is that ‘a word is known by the company it keeps’. Concretely, consider a ‘center’ word c surrounded before and after by a context of a certain length. We term words in this contextual window ‘outside words’ (O). For example, in Figure 1, the context window length is 2, the center word c is ‘banking’, and the outside words are ‘turning’, ‘into’, ‘crises’, and ‘as’:

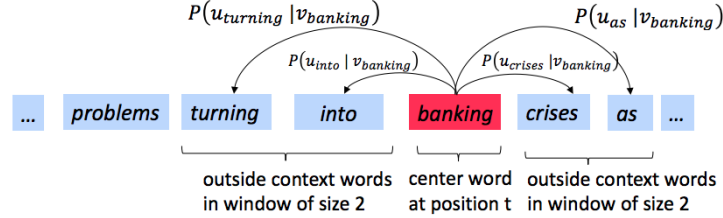


Figure 1: The word2vec skip-gram prediction model with window size 2

Skip-gram word2vec aims to learn the probability distribution $P(O|C)$. Specifically, given a specific word o and a specific word c , we want to predict $P(O = o | C = c)$: the probability that word o is an ‘outside’ word for c (i.e., that it falls within the contextual window of c). We model this probability by taking the softmax function over a series of vector dot-products:

$$P(O = o | C = c) = \frac{\exp(\mathbf{u}_o^\top \mathbf{v}_c)}{\sum_{w \in \text{Vocab}} \exp(\mathbf{u}_w^\top \mathbf{v}_c)} \quad (1)$$

For each word, we learn vectors u and v , where $\mathbf{u}_o$ is the ‘outside’ vector representing outside word o , and $\mathbf{v}_c$ is the ‘center’ vector representing center word c . We store these parameters in two matrices, $\mathbf{U}$ and $\mathbf{V}$. The columns of $\mathbf{U}$ are all the ‘outside’ vectors $\mathbf{u}_w$; the columns of $\mathbf{V}$ are all of the ‘center’ vectors $\mathbf{v}_w$. Both $\mathbf{U}$ and $\mathbf{V}$ contain a vector for every $w \in \text{Vocabulary}$.¹

Recall from lectures that, for a single pair of words c and o , the loss is given by:

$$\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U}) = -\log P(O = o | C = c). \quad (2)$$

¹Assume that every word in our vocabulary is matched to an integer number k . Bolded lowercase letters represent vectors. $\mathbf{u}_k$ is both the k^{th} column of $\mathbf{U}$ and the ‘outside’ word vector for the word indexed by k . $\mathbf{v}_k$ is both the k^{th} column of $\mathbf{V}$ and the ‘center’ word vector for the word indexed by k . **In order to simplify notation we shall interchangeably use k to refer to word k and the index of word k .**

We can view this loss as the cross-entropy² between the true distribution $\mathbf{y}$ and the predicted distribution $\hat{\mathbf{y}}$, for a particular center word c and a particular outside word o . Here, both $\mathbf{y}$ and $\hat{\mathbf{y}}$ are vectors with length equal to the number of words in the vocabulary. Furthermore, the k^{th} entry in these vectors indicates the conditional probability of the k^{th} word being an ‘outside word’ for the given c . The true empirical distribution $\mathbf{y}$ is a one-hot vector with a 1 for the true outside word o , and 0 everywhere else, for this particular example of center word c and outside word o .³ The predicted distribution $\hat{\mathbf{y}}$ is the probability distribution $P(O|C=c)$ given by our model in equation (1).

Note: Throughout this homework, when computing derivatives, please use the method reviewed during the lecture (i.e. no Taylor Series Approximations).

- (a) (2 points) Prove that the naive-softmax loss (Equation 2) is the same as the cross-entropy loss between $\mathbf{y}$ and $\hat{\mathbf{y}}$, i.e. (note that $\mathbf{y}$ (true distribution), $\hat{\mathbf{y}}$ (predicted distribution) are vectors and $\hat{\mathbf{y}}_o$ is a scalar):

$$-\sum_{w \in \text{Vocab}} \mathbf{y}_w \log(\hat{\mathbf{y}}_w) = -\log(\hat{\mathbf{y}}_o). \quad (3)$$

Your answer should be one line. You may describe your answer in words.

- (b) (6 points) i. Compute the partial derivative of $\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U})$ with respect to $\mathbf{v}_c$. *Please write your answer in terms of $\mathbf{y}$, $\hat{\mathbf{y}}$, $\mathbf{U}$, and show your work to receive full credit.*

- **Note:** Your final answers for the partial derivative should follow the shape convention: the partial derivative of any function $f(x)$ with respect to x should have the **same shape** as x .⁴
- Please provide your answers for the partial derivative in vectorized form. For example, when we ask you to write your answers in terms of $\mathbf{y}$, $\hat{\mathbf{y}}$, and $\mathbf{U}$, you may not refer to specific elements of these terms in your final answer (such as $\mathbf{y}_1$, $\mathbf{y}_2$, ...). You may also not refer to specific vectors such as u_0 , u_1 , etc.

- ii. When is the gradient you computed equal to zero? Write a mathematical equation.

Hint: You may wish to review and use some introductory linear algebra concepts.

- iii. The gradient you found is the difference between the two terms. Provide an interpretation of how each of these terms improves the word vector when this gradient is subtracted from the word vector $\mathbf{v}_c$.

- (c) (1 point) In many downstream applications using word embeddings, L2 normalized vectors (e.g. $\mathbf{u}/\|\mathbf{u}\|_2$ where $\|\mathbf{u}\|_2 = \sqrt{\sum_i u_i^2}$) are used instead of their raw forms (e.g. $\mathbf{u}$). Let’s consider a hypothetical downstream task of binary classification of phrases as being positive or negative, where you decide the sign based on the sum of individual embeddings of the words. When would L2 normalization take away useful information for the downstream task? When would it not?

Hint: Consider the case where $\mathbf{u}_x = \alpha \mathbf{u}_y$ for some words $x \neq y$ and some scalar α . When α is positive, what will be the value of normalized $\mathbf{u}_x$ and normalized $\mathbf{u}_y$? How might $\mathbf{u}_x$ and $\mathbf{u}_y$ be related for such a normalization to affect or not affect the resulting classification?

- (d) (1 point) Write down the partial derivative of $\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U})$ with respect to $\mathbf{U}$. Please break down your answer in terms of the column vectors $\frac{\partial \mathbf{J}(\mathbf{v}_c, o, \mathbf{U})}{\partial \mathbf{u}_1}$, $\frac{\partial \mathbf{J}(\mathbf{v}_c, o, \mathbf{U})}{\partial \mathbf{u}_2}$, ..., $\frac{\partial \mathbf{J}(\mathbf{v}_c, o, \mathbf{U})}{\partial \mathbf{u}_{|\text{Vocab}|}}$ (do not further expand these terms). No derivations are necessary, just an answer in the form of a matrix.

²The **cross-entropy loss** between the true (discrete) probability distribution p and another distribution q is $-\sum_i p_i \log(q_i)$.

³Note that the true conditional probability distribution of context words for the entire training dataset would not be one-hot.

⁴This allows us to efficiently minimize a function using gradient descent without worrying about reshaping or dimension mismatching. While following the shape convention, we’re guaranteed that $\theta := \theta - \alpha \frac{\partial J(\theta)}{\partial \theta}$ is a well-defined update rule.

- (e) (5 points) Compute the partial derivatives of $\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U})$ with respect to each of the ‘outside’ word vectors, $\mathbf{u}_w$ ’s. There will be two cases: when $w = o$, the true ‘outside’ word vector, and $w \neq o$, for all other words. Please write your answer in terms of $\mathbf{y}$, $\hat{\mathbf{y}}$, and $\mathbf{v}_c$. In this subpart, you may use specific elements within these terms as well (such as $\mathbf{y}_1, \mathbf{y}_2, \dots$). Note that $\mathbf{u}_w$ is a vector while $\mathbf{y}_1, \mathbf{y}_2, \dots$ are scalars. Show your work to receive full credit.
- (f) (2 points) As an additional exercise for this problem, you will be taking the derivatives of some common loss functions, which may be used in variations of `word2vec` (such as the negative sampling variant). The Leaky ReLU (Leaky Rectified Linear Unit) activation function is given by Equation 4 and Figure 2:

$$f(x) = \max(\alpha x, x) \quad (4)$$

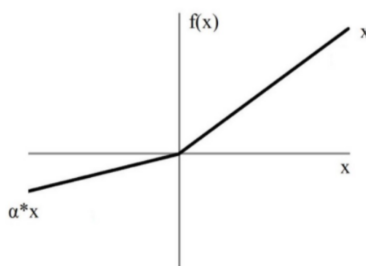


Figure 2: Leaky ReLU

Where x is a scalar and $0 < \alpha < 1$, please compute the derivative of $f(x)$ with respect to x . You may ignore the case where the derivative is not defined at 0.⁵

- (g) (3 points) The sigmoid function is given by Equation 5:

$$\sigma(x) = \frac{1}{1 + e^{-x}} = \frac{e^x}{e^x + 1} \quad (5)$$

Please compute the derivative of $\sigma(x)$ with respect to x , where x is a scalar. Please write your answer in terms of $\sigma(x)$. Show your work to receive full credit.

⁵If you're interested in how to handle the derivative at this point, you can read more about the notion of subderivatives.

2. Neural Networks Optimization (8 points)

- (a) (4 points) Adam Optimizer

Recall the standard Stochastic Gradient Descent update rule:

$$\theta_{t+1} \leftarrow \theta_t - \alpha \nabla_{\theta_t} J_{\text{minibatch}}(\theta_t)$$

where $t + 1$ is the current timestep, θ is a vector containing all of the model parameters, (θ_t is the model parameter at time step t , and θ_{t+1} is the model parameter at time step $t + 1$), J is the loss function, $\nabla_{\theta} J_{\text{minibatch}}(\theta)$ is the gradient of the loss function with respect to the parameters on a minibatch of data, and α is the learning rate. Adam Optimization⁶ uses a more sophisticated update rule with two additional steps.⁷

- i. (2 points) First, Adam uses a trick called *momentum* by keeping track of $\mathbf{m}$, a rolling average of the gradients:

$$\begin{aligned} \mathbf{m}_{t+1} &\leftarrow \beta_1 \mathbf{m}_t + (1 - \beta_1) \nabla_{\theta_t} J_{\text{minibatch}}(\theta_t) \\ \theta_{t+1} &\leftarrow \theta_t - \alpha \mathbf{m}_{t+1} \end{aligned}$$

where β_1 is a hyperparameter between 0 and 1 (often set to 0.9). **Briefly explain in 2–4 sentences** (you don't need to prove mathematically, just give an intuition) how using $\mathbf{m}$ stops the updates from varying as much and why this low variance may be helpful to learning, overall.

- ii. (2 points) Adam extends the idea of *momentum* with the trick of *adaptive learning rates* by keeping track of $\mathbf{v}$, a rolling average of the magnitudes of the gradients:

$$\begin{aligned} \mathbf{m}_{t+1} &\leftarrow \beta_1 \mathbf{m}_t + (1 - \beta_1) \nabla_{\theta_t} J_{\text{minibatch}}(\theta_t) \\ \mathbf{v}_{t+1} &\leftarrow \beta_2 \mathbf{v}_t + (1 - \beta_2) (\nabla_{\theta_t} J_{\text{minibatch}}(\theta_t) \odot \nabla_{\theta_t} J_{\text{minibatch}}(\theta_t)) \\ \theta_{t+1} &\leftarrow \theta_t - \alpha \mathbf{m}_{t+1} / \sqrt{\mathbf{v}_{t+1}} \end{aligned}$$

where $\odot$ and $/$ denote elementwise multiplication and division (so $\mathbf{z} \odot \mathbf{z}$ is elementwise squaring) and β_2 is a hyperparameter between 0 and 1 (often set to 0.99). Since Adam divides the update by $\sqrt{\mathbf{v}}$, which of the model parameters will get larger updates? Why might this help with learning? **Briefly explain in 2–4 sentences.**

- (b) (4 points) Dropout⁸ is a regularization technique. During training, dropout randomly sets units in the hidden layer $\mathbf{h}$ to zero with probability p_{drop} (dropping different units each minibatch), and then multiplies $\mathbf{h}$ by a constant γ . We can write this as:

$$\mathbf{h}_{\text{drop}} = \gamma \mathbf{d} \odot \mathbf{h}$$

where $\mathbf{d} \in \{0, 1\}^{D_h}$ (D_h is the size of $\mathbf{h}$) is a mask vector where each entry is 0 with probability p_{drop} and 1 with probability $(1 - p_{\text{drop}})$. γ is chosen such that the expected value of $\mathbf{h}_{\text{drop}}$ is $\mathbf{h}$:

$$\mathbb{E}_{p_{\text{drop}}}[\mathbf{h}_{\text{drop}}]_i = h_i$$

for all $i \in \{1, \dots, D_h\}$.

- i. (2 points) What must γ equal in terms of p_{drop} ? Briefly justify your answer or show your math derivation using the equations given above.
- ii. (2 points) Why should dropout be applied during training? Why should dropout **NOT** be applied during evaluation? **Briefly explain in 2–4 sentences. Hint:** it may help to look at the dropout paper linked.

⁶Kingma and Ba, 2015, <https://arxiv.org/pdf/1412.6980.pdf>

⁷The actual Adam update uses a few additional tricks that are less important, but we won't worry about them here. If you want to learn more about it, you can take a look at: <http://cs231n.github.io/neural-networks-3/#sgd>

⁸Srivastava et al., 2014, <https://www.cs.toronto.edu/~hinton/absps/JMLRdropout.pdf>

3. Neural Transition-Based Dependency Parsing (40 points)

In this section, you'll be implementing a neural-network based dependency parser with the goal of maximizing performance on the UAS (Unlabeled Attachment Score) metric.

Before you begin, please run `uv sync` and `uv venv` in your terminal inside the unzipped Assignment2 directory. Then select the Python kernel as `.venv`.

A dependency parser analyzes the grammatical structure of a sentence, establishing relationships between *head* words, and words which modify those heads. There are multiple types of dependency parsers, including transition-based parsers, graph-based parsers, and feature-based parsers. Your implementation will be a *transition-based* parser, which incrementally builds up a parse one step at a time. At every step it maintains a *partial parse*, which is represented as follows:

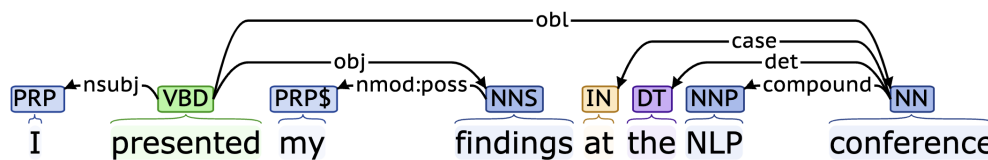
- A *stack* of words that are currently being processed.
- A *buffer* of words yet to be processed.
- A list of *dependencies* predicted by the parser.

Initially, the stack only contains ROOT, the dependencies list is empty, and the buffer contains all words of the sentence in order. At each step, the parser applies a *transition* to the partial parse until its buffer is empty and the stack size is 1. The following transitions can be applied:

- **SHIFT**: removes the first word from the buffer and pushes it onto the stack.
- **LEFT-ARC**: marks the second (second most recently added) item on the stack as a dependent of the first item and removes the second item from the stack, adding a *first_word* $\rightarrow$ *second_word* dependency to the dependency list.
- **RIGHT-ARC**: marks the first (most recently added) item on the stack as a dependent of the second item and removes the first item from the stack, adding a *second_word* $\rightarrow$ *first_word* dependency to the dependency list.

On each step, your parser will decide among the three transitions using a neural network classifier.

- (a) (4 points) Go through the sequence of transitions needed for parsing the sentence “*I presented my findings at the NLP conference*”. The dependency tree for the sentence is shown below. At each step, give the configuration of the stack and buffer, as well as what transition was applied this step and what new dependency was added (if any). The first three steps are provided below as an example.



Stack	Buffer	New dependency	Transition
[ROOT]	[I, presented, my, findings, at, the, NLP, conference]		Initial Configuration
[ROOT, I]	[presented, my, findings, at, the, NLP, conference]		SHIFT
[ROOT, I, presented]	[my, findings, at, the, NLP, conference]		SHIFT
[ROOT, presented]	[my, findings, at, the, NLP, conference]	presented $\rightarrow$ I	LEFT-ARC

- (b) (2 points) A sentence containing n words will be parsed in how many steps (in terms of n)? Briefly explain in 1–2 sentences why.

- (c) (6 points) Implement the `__init__` and `parse_step` functions in the `PartialParse` class in `parser_transitions.py`. This implements the transition mechanics your parser will use. You can run basic (non-exhaustive) tests by running `uv run parser_transitions.py part_c`.
- (d) (8 points) Our network will predict which transition should be applied next to a partial parse. We could use it to parse a single sentence by applying predicted transitions until the parse is complete. However, neural networks run much more efficiently when making predictions about *batches* of data at a time (i.e., predicting the next transition for any different partial parses simultaneously). We can parse sentences in minibatches with the following algorithm.

Algorithm 1 Minibatch Dependency Parsing

Input: `sentences`, a list of sentences to be parsed and `model`, our model that makes parse decisions

Initialize `partial_parses` as a list of `PartialParses`, one for each sentence in `sentences`

Initialize `unfinished_parses` as a shallow copy of `partial_parses`

while `unfinished_parses` is not empty **do**

 Take the first `batch_size` parses in `unfinished_parses` as a minibatch

 Use the `model` to predict the next transition for each partial parse in the minibatch

 Perform a parse step on each partial parse in the minibatch with its predicted transition

 Remove the completed (empty buffer and stack of size 1) parses from `unfinished_parses`

end while

Return: The dependencies for each (now completed) parse in `partial_parses`.

Implement this algorithm in the `minibatch_parse` function in `parser_transitions.py`. You can run basic (non-exhaustive) tests by running `uv run parser_transitions.py part_d`.

Note: You will need `minibatch_parse` to be correctly implemented to evaluate the model you will build in part (e). However, you do not need it to train the model, so you should be able to complete most of part (e) even if `minibatch_parse` is not implemented yet.

- (e) (20 points) We are now going to train a neural network to predict, given the state of the stack, buffer, and dependencies, which transition should be applied next.

First, the model extracts a feature vector representing the current state. We will be using the feature set presented in the original neural dependency parsing paper: *A Fast and Accurate Dependency Parser using Neural Networks*.⁹ The function extracting these features has been implemented for you in `utils/parser_utils.py`. This feature vector consists of a list of tokens (e.g., the last word in the stack, first word in the buffer, dependent of the second-to-last word in the stack if there is one, etc.). They can be represented as a list of integers $\mathbf{w} = [w_1, w_2, \dots, w_m]$ where m is the number of features and each $0 \leq w_i < |V|$ is the index of a token in the vocabulary ($|V|$ is the vocabulary size). Then our network looks up an embedding for each word and concatenates them into a single input vector:

$$\mathbf{x} = [\mathbf{E}_{w_1}, \dots, \mathbf{E}_{w_m}] \in \mathbb{R}^{dm}$$

where $\mathbf{E} \in \mathbb{R}^{|V| \times d}$ is an embedding matrix with each row $\mathbf{E}_w$ as the vector for a particular word w

⁹Chen and Manning, 2014, <https://nlp.stanford.edu/pubs/emnlp2014-depparser.pdf>

with dimension d . We then compute our prediction as:

$$\begin{aligned}\mathbf{h} &= \text{ReLU}(\mathbf{x}\mathbf{W} + \mathbf{b}_1) \\ \mathbf{l} &= \mathbf{h}\mathbf{U} + \mathbf{b}_2 \\ \hat{\mathbf{y}} &= \text{softmax}(\mathbf{l})\end{aligned}$$

where $\mathbf{h}$ is referred to as the hidden layer, $\mathbf{l}$ is referred to as the logits, $\hat{\mathbf{y}}$ is referred to as the predictions, and $\text{ReLU}(z) = \max(z, 0)$. We will train the model to minimize cross-entropy loss:

$$J(\theta) = CE(\mathbf{y}, \hat{\mathbf{y}}) = - \sum_{j=1}^3 y_j \log \hat{y}_j$$

where y_j denotes the j th element of $\mathbf{y}$. To compute the loss for the training set, we average this $J(\theta)$ across all training examples.

- i. Compute the derivative of $\mathbf{h} = \text{ReLU}(\mathbf{x}\mathbf{W} + \mathbf{b}_1)$ with respect to $\mathbf{x}$. For simplicity, you only need to show the derivative $\frac{\partial h_i}{\partial x_j}$ for some index i and j . You may ignore the case where the derivative is not defined at 0.
- ii. Recall in part 1b, we computed the partial derivative of $\mathbf{J}_{\text{naive-softmax}}(\mathbf{v}_c, o, \mathbf{U})$. Likewise, please compute the partial derivative of $J(\theta)$ with respect to the i th entry of $\mathbf{l}$, which is denoted as l_i . Specifically, compute $\frac{\partial CE(\mathbf{y}, \hat{\mathbf{y}})}{\partial l_i}$, assuming that $\mathbf{l} \in \mathbb{R}^3$, $\hat{\mathbf{y}} \in \mathbb{R}^3$, $\mathbf{y} \in \mathbb{R}^3$, and the true label is c (i.e., $y_j = 1$ if $j = c$). **Hint:** Use the chain rule: $\frac{\partial J}{\partial l} = \frac{\partial J}{\partial \hat{\mathbf{y}}} \cdot \frac{\partial \hat{\mathbf{y}}}{\partial \mathbf{l}}$.
- iii. We will use UAS score as our evaluation metric. UAS refers to Unlabeled Attachment Score, which is computed as the ratio between number of correctly predicted dependencies and the number of total dependencies despite of the relations (our model doesn't predict this).

In `parser_model.py` you will find skeleton code to implement this simple neural network using PyTorch. Complete the `__init__`, `embedding_lookup` and `forward` functions to implement the model. Then complete the `train_for_epoch` and `train` functions within the `run.py` file.

Finally execute `uv run run.py` to train your model and compute predictions on test data from Penn Treebank (annotated with Universal Dependencies).

Note:

- For this assignment, you are asked to implement Linear layer and Embedding layer. Please **DO NOT** use `torch.nn.Linear` or `torch.nn.Embedding` module in your code, otherwise you will receive deductions for this problem.
- Please follow the naming requirements in our TODO if there are any, e.g. if there are explicit requirements about variable names you have to follow them in order to receive full credits. You are free to declare other variable names if not explicitly required.

Hints:

- Each of the variables you are asked to declare (`self.embed_to_hidden_weight`, `self.embed_to_hidden_bias`, `self.hidden_to_logits_weight`, `self.hidden_to_logits_bias`) corresponds to one of the variables above ($\mathbf{W}$, $\mathbf{b}_1$, $\mathbf{U}$, $\mathbf{b}_2$).
- It may help to work backwards in the algorithm (start from $\hat{\mathbf{y}}$) and keep track of the matrix/vector sizes.
- Once you have implemented `embedding_lookup` (e) or `forward` (f) you can call `uv run parser_model.py` with flag `-e` or `-f` or both to run sanity checks with each

function. These sanity checks are fairly basic and passing them doesn't mean your code is bug free.

- When debugging, you can add a debug flag: `uv run run.py -d`. This will cause the code to run over a small subset of the data, so that training the model won't take as long. Make sure to remove the `-d` flag to run the full model once you are done debugging.
- When running with debug mode, you should be able to get a loss smaller than 0.2 and a UAS larger than 65 on the dev set (although in rare cases your results may be lower, there is some randomness when training).
- It should take up to **15 minutes** to train the model on the entire training dataset, i.e., when debug mode is disabled.
- When debug mode is disabled, you should be able to get a loss smaller than 0.08 on the train set and an Unlabeled Attachment Score larger than 87 on the dev set. For comparison, the model in the original neural dependency parsing paper gets 92.5 UAS. If you want, you can tweak the hyperparameters for your model (hidden layer size, hyperparameters for Adam, number of epochs, etc.) to improve the performance (but you are not required to do so).

Deliverables:

- Working implementation of the transition mechanics that the neural dependency parser uses in `parser_transitions.py`.
- Working implementation of minibatch dependency parsing in `parser_transitions.py`.
- Working implementation of the neural dependency parser in `parser_model.py`. (We'll look at and run this code for grading).
- Working implementation of the functions for training in `run.py`. (We'll look at and run this code for grading).
- **Report the best UAS your model achieves on the dev set and the UAS it achieves on the test set in your written submission.** You can report it in the PDF and tag the page.

Submission Instructions

For the written question write your answers with any software that can export a pdf file. Zip `parser_model.py`, `parser_transitions.py` and `run.py` files and your pdf file into the zip file named `StudentID_YourName.zip`. Submit the zip file through LMS